take ideas that begin as speculation, and prove them wrong or transform them into educated guesses or firm science.

I do this in two ways: First, I *explain* what is known today about phenomena seen in the movie (black holes, wormholes, singularities, the fifth dimension, and the like), and I explain how we learned what we know, and how we hope to master the unknown. Second, I *interpret*, from a scientist's viewpoint, what we see in *Interstellar*, much as an art critic or ordinary viewer interprets a Picasso painting.

My interpretation is often a description of what I imagine might be going on behind the scenes: the physics of the black hole Gargantua, its singularities, horizon, and visual appearance; how Gargantua's tidal gravity could generate 4000-foot water waves on Miller's planet; how the tesseract, an object with four space dimensions, could transport three-dimensional Cooper through the five-dimensional bulk; . . .

Sometimes my interpretation is an *extrapolation* of *Interstellar*'s story beyond what we see in the movie; for example, how Professor Brand, long before the movie began, might have discovered the wormhole, via gravitational waves that traveled from a neutron star near Gargantua through the wormhole to Earth.

These interpretations, of course, are my own. They are not endorsed by Christopher Nolan any more than an art critic's interpretations were endorsed by Pablo Picasso. They are my vehicle for describing some wonderful science.

Some segments of this book may be rough going. That's the nature of real science. It requires thought. Sometimes deep thought. But thinking can be rewarding. You can just skip the rough parts, or you can struggle to understand. If your struggle is fruitless, then that's my fault, not yours, and I apologize.

I hope that at least once you find yourself, in the dead of night, half asleep, puzzling over something I have written, as I puzzled at night